Quantification: 877_wt_z087c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	9
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3532.86	3114	2023.13	px^2
major_axis_length_px	lacuna shape	98.6806	93.6668	25.9271	px
minor_axis_length_px	lacuna shape	46.6071	46.7845	15.9389	px
aspect_ratio	lacuna shape	2.2368	2.0448	0.5814	unitless
eccentricity	lacuna shape	0.8771	0.8723	0.0517	unitless
solidity	lacuna shape	0.7984	0.8167	0.0754	unitless
roots_count	headline	10.1111	12	7.0257	count
ring_length_r30_px	headline	338.778	374	222.312	px
ring_length_r60_px	reported	837	1029	484.304	px
owned_length_px	ownership-dependent	1742.03	1888.16	1287.1	px
edge_count	ownership-dependent	83.5556	97	63.6457	count
mean_edge_length_px	ownership-dependent	21.3672	20.5662	2.7665	px
perimeter_px	normalised	295.653	301.404	104.145	px
ring_area_r30_px2	normalised	9718.33	9757	3317.86	px^2
ring_area_r60_px2	normalised	23702.2	24715	7555.84	px^2
in_frame_fraction_r30	normalised	0.9994	1	0.0019	unitless
in_frame_fraction_r60	normalised	0.9845	1	0.0465	unitless
ring_density_r30	normalised	0.0328836	0.0377938	0.0139814	px^-1
ring_density_r60	normalised	0.0342513	0.0389001	0.0151736	px^-1
roots_per_100px_perimeter	normalised	3.254	3.8176	1.8913	per 100 px
ring_attached_length_r30_px	network v2	300.778	332	221.798	px
ring_attached_length_r60_px	network v2	686	869	466.734	px
ring_length_w_r30_px	network v2	390.454	422.777	256.935	px
ring_length_w_r60_px	network v2	962.124	1160.41	561.734	px
sholl_crossings_r10	network v2	8.1111	9	5.3489	count
sholl_crossings_r20	network v2	10.7778	10	7.0671	count
sholl_crossings_r30	network v2	12.4444	14	6.8577	count
loop_count	loops, report only	7.6667	11	6.892	count
loop_edge_share	loops, report only	0.2259	0.2887	0.1657	share
ring_width_mean_r30_px	width	6.8548	7.0839	0.8622	px

Per field
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measure	value	units
analysed_area_px2	1.009e+06	px^2
total_skeleton_length_px	30983	px
canalicular_length_density_per_px	0.0307068	px^-1
junction_count	631	count
junction_density_per_px2	0.00062537	px^-2
lacuna_count	13	count

lacunae_per_px2	1.288e-05	px^-2
skeleton_component_count	389	count
mean_component_length_px	79.6478	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0351053	px^-1
field_density_without_flagged_per_px	0.0307068	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	716	px
unattached_component_count	36	count
canalicular_length_density_attached_per_px	0.0307068	px^-1
field_length_density_with_unattached_per_px	0.0314164	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	88	count
loop_edge_share	0.1949	share
edge_count	1616	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3241	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	30983	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1757	77.0533	35.9762	2.1418	0.8843	0.7142	1.417	13.6949	531.608	True
2	4317	116.364	50.6241	2.2986	0.9004	0.8545	0.8274	37.7415	814.323	True
3	1047	73.2742	21.411	3.4223	0.9564	0.675	0.4727	102.495	237.658	False
4	5980	133.077	65.0796	2.0448	0.8723	0.7429	0.7964	208.483	500.988	False
5	4142	119.005	48.2749	2.4652	0.914	0.8826	0.2092	399.882	20.3518	True
6	4805	118.959	55.0153	2.1623	0.8866	0.8357	0.4696	413.392	307.567	False
7	3114	93.6668	46.7845	2.0021	0.8663	0.8167	0.8128	455.325	614.1	False
8	5891	121.42	63.3439	1.9168	0.8531	0.8972	1.4767	461.976	812.6	False
9	1692	71.502	34.0722	2.0985	0.8792	0.9082	0.0325	636.572	12.6684	True
10	1983	66.6089	40.8008	1.6325	0.7904	0.8555	1.3986	723.423	816.988	False
11	1910	83.7583	33.8144	2.477	0.9149	0.7658	1.3629	757.078	851.197	False
12								860.708	550.499	False
13								970.308	685.063	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	343	732	1267.79	70	18.1112	7.5173
2	8	245	592	942.034	39	24.1547	7.215
3	13	374	1029	1888.16	97	19.4656	8.1759
4	23	761	1623	3410.19	164	20.7938	7.6347

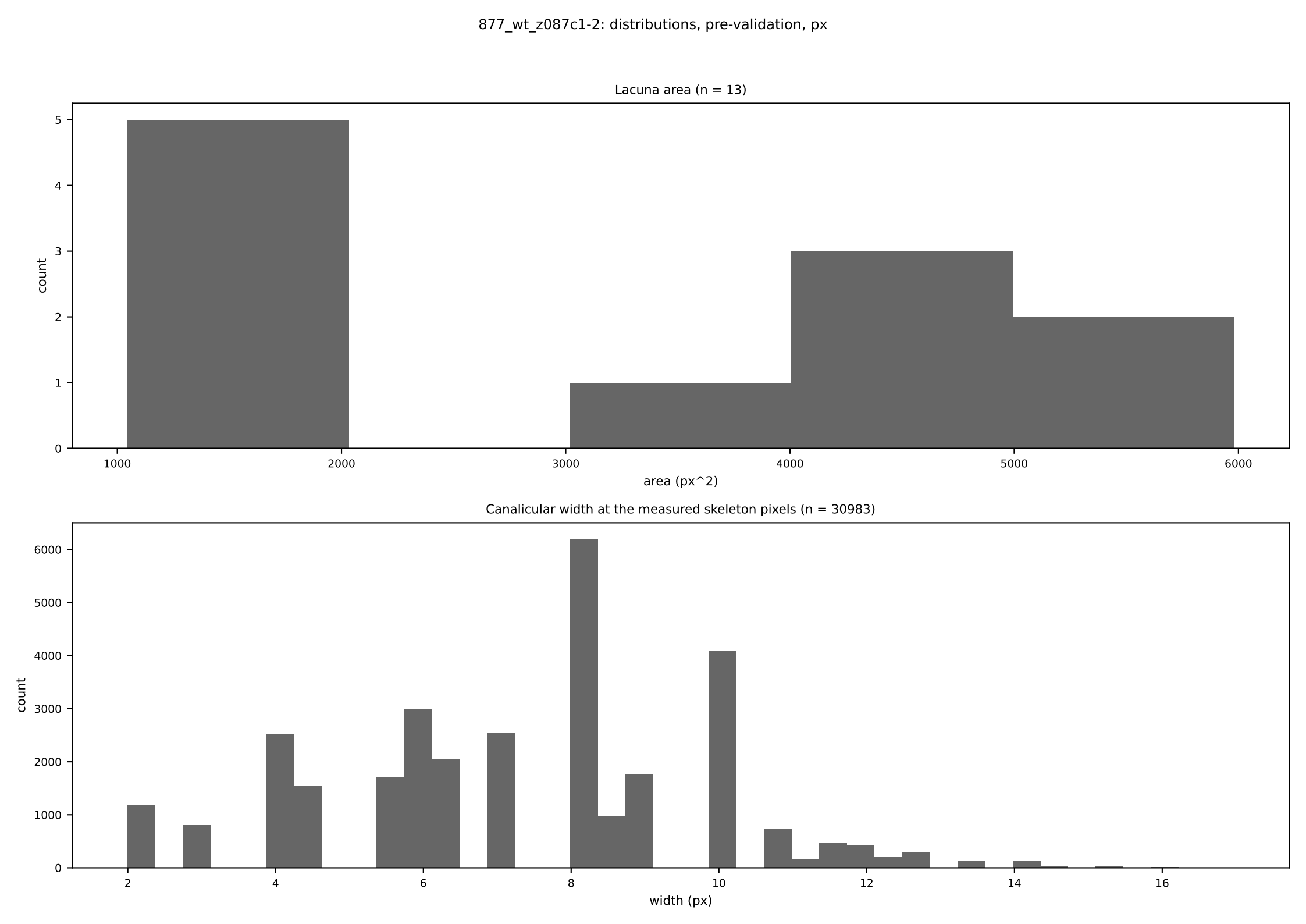
5	5	215	584	438.262	28	15.6522	6.7369
6	13	468	1125	2341.59	125	18.7327	7.2639
7	12	394	1120	2903.15	142	20.4447	6.7385
8	14	494	1090	3040.56	140	21.7183	7.0839
9	6	136	381	316.865	13	24.3742	7.9135
10	5	183	539	593.558	22	26.9799	5.9628
11	9	205	621	1358.74	55	24.7043	7.0844
12	1	108	315	123.397	6	20.5662	6.4177
13	1	62	71	18.8995	1	18.8995	5.3312

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	274.066	6033	14351	0.6294	0.5895	0.056854	0.0510069	3.2839
2	340.475	9033	20414	0.7674	0.7052	0.0271228	0.0289997	2.3497
3	206.551	8300	22062	1	1	0.0450602	0.0466413	6.2938
4	502.186	15464	35653	1	1	0.0492111	0.0455221	4.58
5	336.628	7317	17241	0.6107	0.5856	0.0293836	0.0338728	1.4853
6	357.404	12357	30057	1	1	0.0378733	0.0374289	3.6373
7	314.333	10425	25683	1	1	0.0377938	0.0436086	3.8176
8	364.718	12512	29886	1	1	0.0394821	0.0364719	3.8386
9	190.953	4836	12429	0.5823	0.5596	0.0281224	0.0306541	3.1421
10	215.522	5675	13856	1	1	0.0322467	0.0389001	2.32
11	232.936	5964	14531	1	1	0.0343729	0.0427362	3.8637
12	301.404	9757	24715	1	1	0.011069	0.0127453	0.3318
13	165.823	7011	16877	0.9943	0.8606	0.00884325	0.00420691	0.6031

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	311	618	395.363	846.252	8	7	10
2	227	455	289.492	684.896	8	9	11
3	349	965	422.777	1160.41	9	10	14
4	734	1490	878.394	1882.16	16	21	22
5	153	321	240.823	658.742	6	6	7
6	413	930	554.441	1309.78	13	18	19
7	332	874	448.99	1289.76	10	11	16
8	448	869	565.943	1249.63	13	19	18
9	134	281	154.711	424.848	5	4	7
10	169	406	208.51	618.571	4	5	6
11	181	516	235.309	713.37	5	6	9
12	64	107	130.024	358.333	2	6	4
13	17	17	69.6985	77.1127	1	1	4


877_wt_z087c1-2: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 30983)
count
width (px)